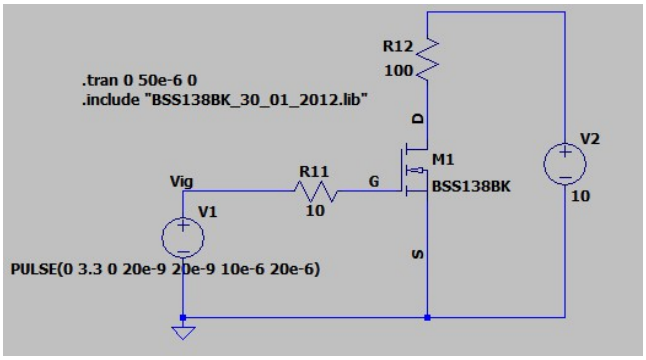
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Taking into account the circuit (below) and the transistor datasheet, analyze the following points:

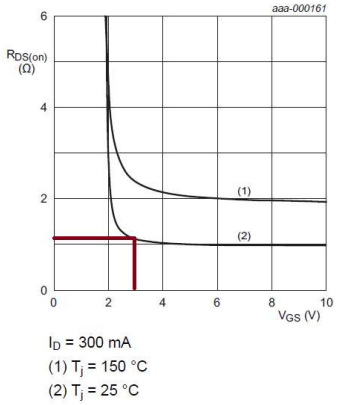


Considering V2 = 10V and Rth(j-a), calculate the conduction losses and the transistor's temperature increase.

Conduction losses calculation

$V2 := 10\ V$ $VGS := 3.3\ V$ $RDSon := 1.1\ ohm$ $on_duty_cycle := 50\ \%$

Fig 9. Drain-source on-state resistance as a function of gate-source voltage; typical values



I_D = 300 mA
(1) T_j = 150 °C
(2) T_j = 25 °C

$R12 := 100\ ohm$

$$IDon := \frac{V2}{R12 + RDSon} = 0.0989\ A$$

$$Pon := IDon^{2} \cdot RDSon \cdot on_duty_cycle = 0.0054\ W$$

Transistor's temperature increase

$Rthja := 310\ \frac{K}{W}$

Table 6. Thermal characteristics

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
$R_{th(j-a)}$	thermal resistance from junction to ambient	in free air	[1]	-	310	370	K/W
			[2]	-	260	300	K/W

$\Delta T := Pon \cdot Rthja = 1.6681\ K$

Simulation results:
vds_on: V(d) =0.117455715582 at 5e-06
id_on: abs(I(R12)) =0.0988254440605 at 5e-06
rds_on: VDS_on/ID_on=1.18851695227
*p_on: 0.5*VDS_on*ID_on=0.00580380662491*
*delta_tj: P_on*310=1.79918005372*

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Considering the circuit as shown above and Application Note AN-1001 (section 5.1), measure (in a simulator) td(on), td(off), tr, and tf of the transistor. Do the measured values match the datasheet? What do you think is the reason for any differences?

7. Characteristics

Table 7. Characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
Static characteristics						
$V_{(BR)DSS}$	drain-source breakdown voltage	I_D = 250 µA; V_{GS} = 0 V; T_j = 25 °C	60	-	-	V
V_{GSth}	gate-source threshold voltage	I_D = 250 µA; V_{DS} = V_{GS}; T_j = 25 °C	0.48	1.1	1.6	V
I_{DSS}	drain leakage current	V_{DS} = 60 V; V_{GS} = 0 V; T_j = 25 °C	-	-	1	µA
		V_{DS} = 60 V; V_{GS} = 0 V; T_j = 150 °C	-	-	10	µA
I_{GSS}	gate leakage current	V_{GS} = 20 V; V_{DS} = 0 V; T_j = 25 °C	-	-	10	µA
		V_{GS} = -20 V; V_{DS} = 0 V; T_j = 25 °C	-	-	10	µA
		V_{GS} = 10 V; V_{DS} = 0 V; T_j = 25 °C	-	-	1	µA
		V_{GS} = -10 V; V_{DS} = 0 V; T_j = 25 °C	-	-	1	µA
R_{DSon}	drain-source on-state resistance	V_{GS} = 10 V; I_D = 350 mA; T_j = 25 °C	-	1	1.6	Ω
		V_{GS} = 10 V; I_D = 350 mA; T_j = 150 °C	-	2	3.2	Ω
		V_{GS} = 4.5 V; I_D = 200 mA; T_j = 25 °C	-	1.1	2.2	Ω
		V_{GS} = 2.5 V; I_D = 10 mA; T_j = 25 °C	-	1.4	6.5	Ω
g_{fs}	forward transconductance	V_{DS} = 10 V; I_D = 200 mA; T_j = 25 °C	-	700	-	mS
Dynamic characteristics						
$Q_{G(tot)}$	total gate charge	V_{DS} = 30 V; I_D = 300 mA; V_{GS} = 4.5 V; T_j = 25 °C	-	0.6	0.7	nC
Q_{GS}	gate-source charge		-	0.1	-	nC
Q_{GD}	gate-drain charge		-	0.2	-	nC
C_{iss}	input capacitance	V_{DS} = 10 V; f = 1 MHz; V_{GS} = 0 V; T_j = 25 °C	-	42	56	pF
C_{oss}	output capacitance		-	7	-	pF
C_{rss}	reverse transfer capacitance		-	4	-	pF
$t_{d(on)}$	turn-on delay time	V_{DS} = 40 V; R_L = 250 Ω; V_{GS} = 10 V; $R_{G(ext)}$ = 6 Ω; T_j = 25 °C	-	5	10	ns
t_r	rise time		-	5	-	ns
$t_{d(off)}$	turn-off delay time		-	38	76	ns
t_f	fall time		-	20	-	ns
Source-drain diode						
V_{SD}	source-drain voltage	I_S = 300 mA; V_{GS} = 0 V; T_j = 25 °C	0.47	0.8	1.2	V

Simulation results:

td_on=7.37073677501e-09 FROM 2.36568361679e-09 TO 9.7364203918e-09
tr=5.1889222348e-09 FROM 9.7364203918e-09 TO 1.49253426266e-08
td_off=1.18515528854e-08 FROM 1.0022476336e-05 TO 1.00343278889e-05
tf=3.9773862807e-09 FROM 1.00343278889e-05 TO 1.00383052752e-05

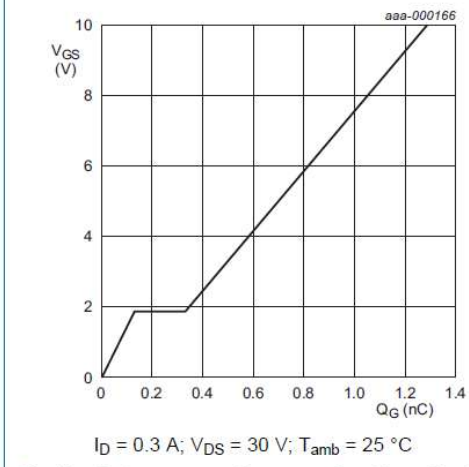
Do the measured values match the datasheet? What do you think is the reason for any differences?

The simulated rise time was close to the typical datasheet value, while the turn-on delay was somewhat longer. Both the turn-off delay and the fall time were substantially shorter than the datasheet values. The test conditions likely contributed to these differences. Our circuit drove the gate to 3.3 V, compared with 10 V in the datasheet test, so less gate charge had to be removed before began to rise. This helps explain the shorter turn-off delay. The drain supply voltage, load resistance, and external gate resistance also differed and can affect the fall time.

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Change the value of R11 to 1kOhm. Considering Application Note AN-1001 (page 14 of the PDF) and the gate charging operations, can you identify the plateau (Vp in the application note)? What causes this plateau? What happens to the gate input current? How can you reduce this plateau?



I_D = 0.3 A; V_{DS} = 30 V; T_{amb} = 25 °C

Fig 14. Gate-source voltage as a function of gate charge; typical values

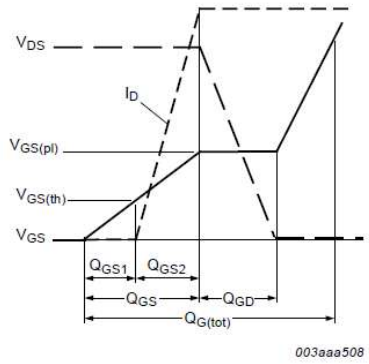
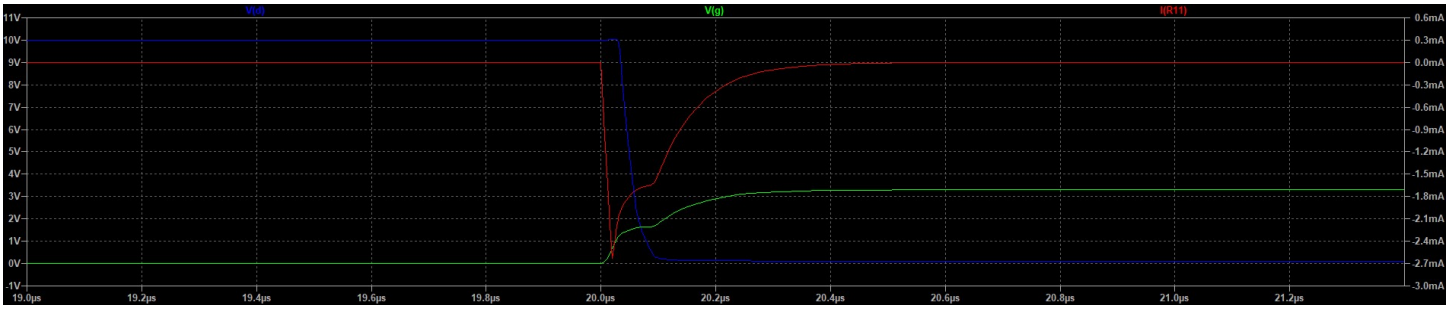


Fig 15. Gate charge waveform definitions



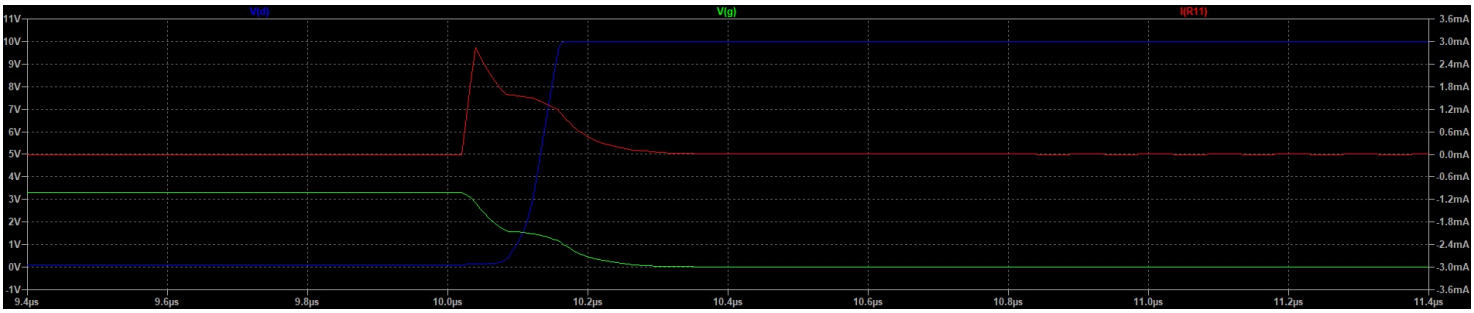
What causes this plateau?
The plateau is caused by the Miller effect. As VDS falls, much of the gate current goes into changing the charge on the gate–drain capacitance. As a result, VGS rises only slightly during this interval.

What happens to the gate input current?
The gate current rises rapidly at turn-on, then stays at roughly 1.8mA in magnitude during the plateau. After the plateau, it approaches zero as VGS approaches 3.3V.

How can you reduce this plateau?
Reducing R11 from 1kOhm to the original 10Ohm allows more current to flow into the gate. This transfers the gate–drain charge more quickly and shortens the plateau.

Can the turn-on and turn-off times of the transistor be reduced?

Yes. Both the turn-on and turn-off times can be reduced by charging and discharging the gate more quickly. In this circuit, reducing R11 from 1kOhm to 10Ohm increased the gate current during both transitions and shortens the Miller plateau.



https://github.com/carloszlopez/Analog-design-class/tree/Assynchronous_Class_Deliverable_6

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